

Q What are the characteristics of Turner's syndrome, and can it be treated?

This rare chromosomal condition affects only girls and occurs when there is one X chromosome missing from the pair that determines her gender. It affects about one in 2,000 baby girls born in the US every year and is named after the American doctor Henry Turner, who first described it in 1938. One of the main characteristics of the condition is short stature, and most girls with Turner's syndrome are also infertile. The condition results in several other distinctive physical features, including a broad neck, ears that are set low on the sides of the head, nipples that appear far apart, a low hairline, and swelling in the hands and feet. Other problems may include hearing problems, short fingers, thyroid problems, and diabetes.

Although there is no specific treatment for Turner's syndrome, growth-hormone supplements and supplements of estrogen can help to stimulate growth and bring on puberty, though most girls still remain infertile. The side effects of the condition are treated on a case-by-case basis.

Q What causes the condition congenital hypothyroidism?

Congenital hypothyroidism (CH) is a rare condition, occurring in only one in every 2,370 births in the US each year and is slightly more common in girls than in boys. It is thought to be caused by malformations in the thyroid gland during development in the uterus (usually because the gland doesn't move into a forward position in the throat), or perhaps the absence of the thyroid gland altogether. Occasionally, an inherited gene mutation may be the cause. The condition means that babies are born without the ability to make thyroxine, an important growth hormone, and if left untreated, hypothyroidism can lead to long-term growth problems and neurological damage.

Your baby will be tested for levels of thyroxine as part of the standard heel-prick test given at five days old. If your baby has CH, she will be given thyroxine supplements that she will need to continue throughout life. As long as the dosage remains accurate, the condition has no long-term consequences for your baby.

Q What is neurofibromatosis?

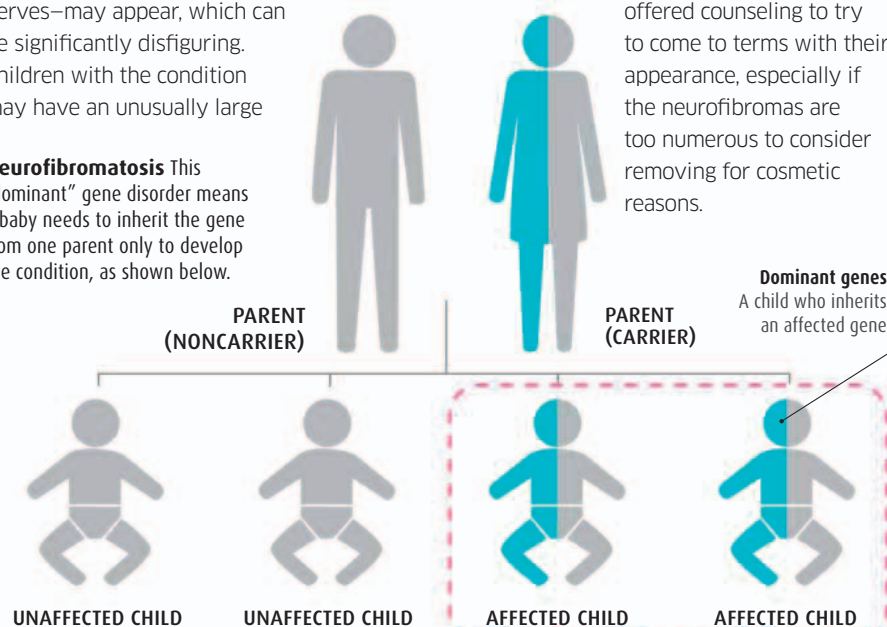
This rare genetic condition affects the nervous system, causing noncancerous tumors to grow on and along the nerves. The condition has a range of symptoms, (and is also associated with learning difficulties).

Babies may have clusters of freckles around the groin and armpits, and flat, light-brown pigmented patches of skin elsewhere on the body. Sometimes freckles appear on the whites of the eyes, and benign tumors may occur on the optic nerve. All over the skin, along nerve pathways, small lumps—as a result of the neurofibromas growing on the nerves—may appear, which can be significantly disfiguring. Children with the condition may have an unusually large

head, and may develop convulsions, learning or behavioral problems, and hearing and sight may be affected. There is no treatment for this progressive disorder, although doctors may use chemotherapy or surgery to remove tumors that become cancerous, as well as surgery to remove benign tumors if these start to affect bodily function. Sufferers are

offered counseling to try to come to terms with their appearance, especially if the neurofibromas are too numerous to consider removing for cosmetic reasons.

Neurofibromatosis This "dominant" gene disorder means a baby needs to inherit the gene from one parent only to develop the condition, as shown below.



Q Is sickle-cell disease the same as sickle-cell anemia?

Sickle-cell disease is the umbrella term for a group of conditions that includes sickle-cell anemia. It is an inherited blood disorder in which red blood cells are shaped like a crescent moon—or a sickle. Healthy red blood cells are disc-shaped and bendy, enabling them to pass freely through the body's network of veins and arteries. Conversely, sickle cells are rigid and sticky, which means they become trapped in the junctions between blood vessels. This causes blockages in the circulation, starving bones, muscles, and organs of the oxygen and nutrients they need to grow and develop. In

turn, organ damage, low growth rate, and pain in the joints and muscles occurs. A person with sickle-cell disease will go on to develop sickle-cell anemia, because sickle cells have a short life span and they die off faster than bone marrow can manufacture new cells to replace them. The result is anemia, which is characterized by a pale complexion and causes extreme fatigue.

Your baby is screened for sickle-cell anemia as part of the heel-prick test at five days old. There is no certain cure for sickle-cell disease, though occasionally bone marrow and stem cell transplants can work. Nonetheless, there have been major advances in the understanding of how to treat the symptoms and to minimize its effects, including prolonging life expectancy.